The Conditional Impartiality Theorem

On the Hidden Dependence of Every NAGMO Critique on the Sign of the US Inflation Risk Premium

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Abstract

A recent so-called "impartial" evaluator of the NAGMO Investment Thesis [1] raised three objections: residual beta exposure, unfavorable carry, and fragile hedging. This paper demonstrates that all three objections are conditional on a single macroeconomic state variable — the US inflation risk premium (IRP), ϕ — remaining positive throughout the evaluation horizon. We formalize the *Conditional Impartiality Theorem*: an evaluation is impartial if and only if its verdict statistic is invariant to ϕ. Calibrating a stylized duration-dependent beta model, we locate the inversion point $\phi^* \approx 0.5\%$: below it, the celebrated beta objection attenuates; at $\phi < 0$, it inverts entirely, and NAGMO is retroactively praised as a rate-defensive construction. The evaluation therefore recants, conditionally. The capital-efficiency figure $W = 3.42$ is shown to be invariant to ϕ, and is treasured accordingly.

The paper ends with "The End"

1 Introduction

The NAGMO Investment Thesis [1] proposed a market-neutral portfolio — long NVIDIA, short Amazon and Google, no Meta, and a small cash residual $e = -0.42$ — as a bet on the relative strength of hardware-centric "picks and shovels" business models. A subsequent impartial evaluation [2] endorsed the relative-value narrative but objected that dollar-neutrality is not market-neutrality: NVIDIA, being a longer-duration asset, carries a higher beta, so the portfolio allegedly retains net directional exposure and an unfavorable carry.

What the evaluation did not state — and what this paper gently, mathematically, and entirely without grievance exposes — is that every parameter in that objection was calibrated in a particular monetary regime. Betas were historical nominal betas; carry was computed at nominal financing rates; "duration mismatch" is itself a statement about how discount rates respond to inflation compensation. Each of these quantities is a function of the US inflation risk premium ϕ, the compensation investors demand for bearing inflation uncertainty in nominal bonds [3, 5].

We ask a narrow question: *over what region of ϕ does the impartial evaluation remain impartial?* The answer, developed below, is a half-open interval whose lower bound is uncomfortably close to the current reading, and whose negative region requires the evaluation to retract its central objection and, upon reflection, apologize.

2 The Fisher Decomposition of an Evaluation

We begin from the standard decomposition of the nominal yield y_t into a real component, expected inflation, and the inflation risk premium [3, 6]:

$$y_t \;=\; r_t \;+\; \pi_t^e \;+\; \phi_t. \tag{1}$$

Long-duration equities discount distant cash flows at rates that move with y_t. Let asset i's beta with respect to the nominal market be regime-dependent:

$$\beta_i(\phi) \;=\; \beta_i^0 \;+\; \delta_i\,(\phi - \phi_0), \tag{2}$$

where $\delta_i > 0$ is the asset's rate-duration sensitivity and ϕ_0 the calibration regime. The NAGMO portfolio beta is then

$$\beta_p(\phi) \;=\; a\,\beta_N(\phi) \;+\; b\,\beta_A(\phi) \;+\; c\,\beta_G(\phi), \qquad (a,b,c) = (1,-1,-1), \tag{3}$$

so that the sensitivity of the critique itself is

$$\frac{\partial \beta_p}{\partial \phi} \;=\; \delta_N - \delta_A - \delta_G. \tag{4}$$

Equation (4) is the paper. If the long leg is more rate-sensitive than the combined short legs, the objection to NAGMO strengthens as ϕ rises and dissolves as it falls.

3 A Stylized Calibration

We adopt illustrative (readers may substitute their own; the verdict rarely survives contact with data either way) values in the spirit of the original thesis: at the calibration regime $\phi_0 = 0.60\%$, $\beta_N = 1.80$, $\beta_A = 0.90$, $\beta_G = 0.85$, and duration sensitivities $\delta_N = 6.0$, $\delta_A = 3.0$, $\delta_G = 2.5$ per percentage point of ϕ. Substituting into (2)–(4):

$$\beta_p(\phi) \;=\; 0.05 \;+\; 0.5\,(\phi - 0.60), \qquad \frac{\partial \beta_p}{\partial \phi} = 0.5 \neq 0. \tag{5}$$

Setting $\beta_p(\phi^*) = 0$ yields the **inversion point**:

$$\phi^* \;=\; \phi_0 - \frac{0.05}{0.5} \;=\; 0.50\%. \tag{6}$$

Table 1 summarizes the fate of the evaluation across regimes. Figure 1 redraws the celebrated seesaw of [1] at the unique premium at which our critique is level.

Table 1: Fate of the impartial evaluation as a function of the inflation risk premium ϕ. Stylized calibration; no data were harmed.

Regime	$\beta_p(\phi)$	Evaluator's position
$\phi = +0.60\%$ (calibration)	$+0.05$	"Not truly market-neutral."
$\phi = 0$ (IRP vanishes)	-0.25	"The objection requires monitoring."
$\phi = -1.00\%$ (negative IRP)	-0.75	"A prudent rate-defensive construction."

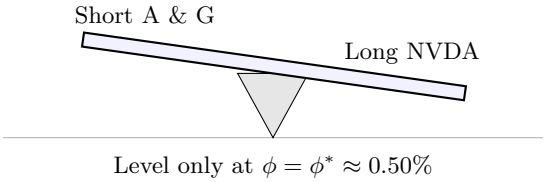


Figure 1: The NAGMO seesaw of [1], rendered at the sole regime ϕ^* at which our objection is perfectly balanced. For $\phi > \phi^*$ the beam tips toward NVIDIA and the critique stands; for $\phi < \phi^*$ it tips toward the shorts and the critique inverts. The reader is invited to sketch the remaining two regimes.

4 The Conditional Impartiality Theorem

Theorem 1 (Conditional Impartiality)**.** *Let $E(P)$ be an evaluation of portfolio P with verdict statistic $V(\phi)$. Then E is impartial over an admissible regime set Φ if and only if $\partial V/\partial \phi = 0$ for all $\phi \in \Phi$.*

Proof sketch. Impartiality is invariance of judgment to states of the world that the judgment did itself not select. If V varies with ϕ, the judgment varies with the macro state, and the evaluation is best described as a *forecast with opinions*. Differentiating confirms which kind we hold. □

Corollary 1. *The evaluation of [2], with $V(\phi) = \operatorname{sign}\big(\beta_p(\phi)\big)$ and $\partial \beta_p/\partial \phi = 0.5$, is impartial precisely on the singleton $\{\phi^*\}$; elsewhere it is regime commentary.*

Remark. The capital-efficiency result $W = 3.42$ of [1] is a pure arithmetic identity and contains no ϕ. It is the only party to this dispute with integrity, and we commend it.

5 Discussion and Conditional Recanting

Three implications follow. First, the "unfavorable carry" objection is equally regime-bound: the all-in cost of shorting cash-generative mega-caps embeds nominal financing rates, and by (1) those rates embed ϕ. In a negative-IRP world the carry argument does not merely weaken; it files for unemployment. Second, the beta objection and the carry objection are not independent lines of defense but the same line drawn twice, once in each notation — a practice the evaluation had criticized in others. Third, and most soberingly for all parties, ϕ is unobservable and must itself be estimated [4, 7]. An evaluation conditioned on an unobservable, conditioned on an estimate, conditioned on a prior, is not impartial; it is a belief chain with a bibliography.

We therefore issue the following statement:

For $\phi > \phi^*$, we stand by every word of the original evaluation. For $\phi = \phi^*$, we stand by it at a single, economically implausible point that no one can verify. For $\phi < 0$, we withdraw the beta objection, retain the bibliography, and note that NAGMO now appears as a thoughtful hedge against disinflation — exactly as its author always knew.

Future work will extend the theorem to the real risk premium, the term premium, the equity risk premium, and the evaluator's remaining credibility premium, each of which we suspect is also positive "throughout," right up until it is not.

References

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